

Sample End Semester Question Paper - 2025

Subject: Engineering Mathematics III | Code: MA301 | Semester: 3

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State the Cauchy-Riemann equations.
2. What is a linear transformation?
3. Define curl.
4. State Stokes' theorem briefly.
5. What is a diagonal matrix?
6. What is an eigenvalue?
7. State one property of rank of a matrix.
8. Define divergence.
9. State Green's theorem briefly.
10. What is Cramer's rule?

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain linear transformations, kernel and image with examples.
2. Explain analytic functions and the Cauchy-Riemann equations with examples.
3. Evaluate a contour integral using Cauchy's theorem and integral formula.
4. Solve a system of linear equations using Cramer's rule.
5. Find the eigenvalues and eigenvectors of a given matrix and diagonalise it.
6. Solve a system of linear equations using matrix methods (Gauss elimination).
7. Explain vector calculus: divergence, curl and gradient with identities.
8. State and apply Green's, Stokes' and the Divergence theorems with examples.